

1. Problems of training your own model?

- ① lot of data
- ② lot of time

gpu

costly

no specific way

- ② How many kernels?
shape of kernels?
what activation
max pool

experimented on lot of data
lot of tuning

3x3

might

① Image net dataset

1.4 million images (1.6m 40k each)

1000 classes

1

build PL (CNN)

best accuracy

TS2V8

97%

VGG16

best

model

tweak

emotion detect

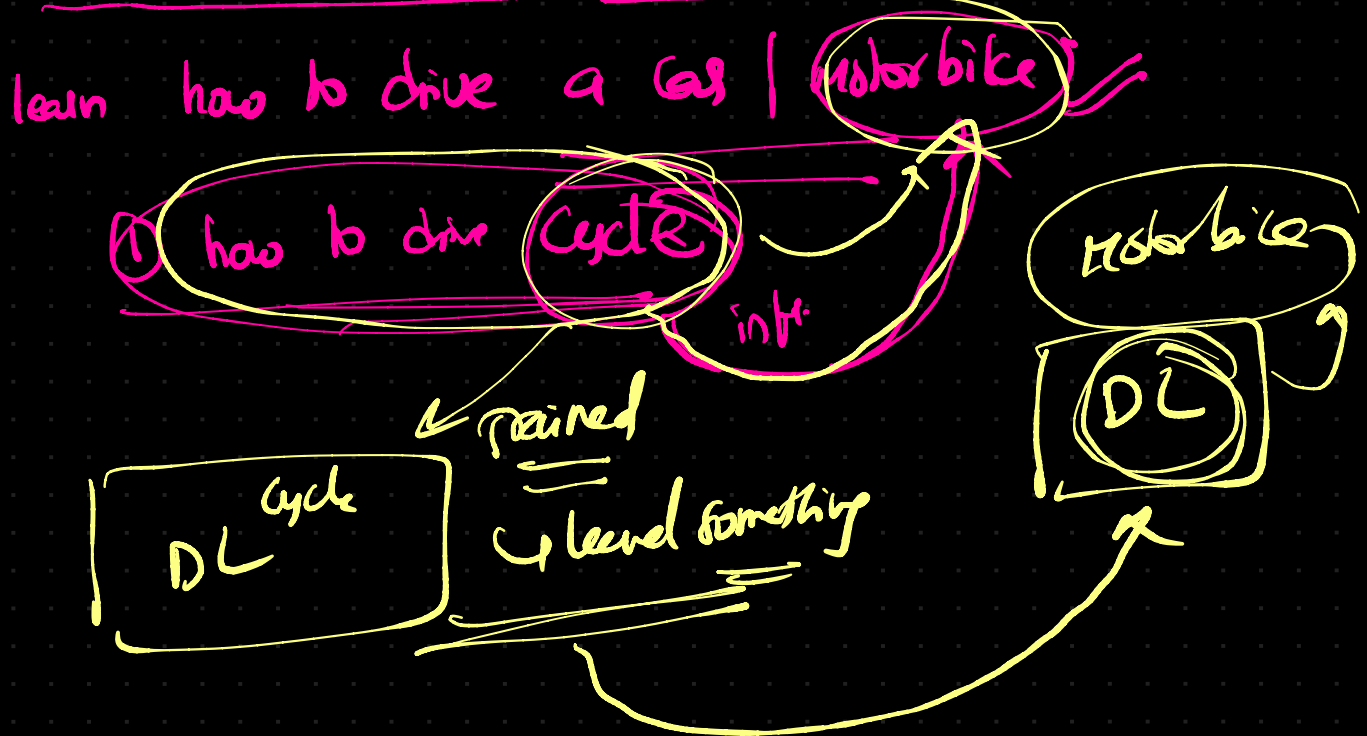
7 classes

VGG16

1000 classes

What is transfer Learning?

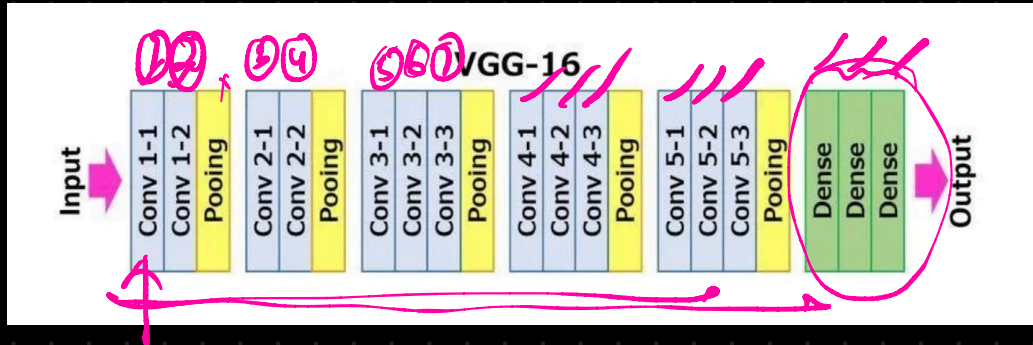
Is a technique that focusses on storing knowledge gained while solving one problem and applying to a different problem.



Note Book Link: <https://colab.research.google.com/drive/1w4n9NLAjwjxtYsE5zkhT7pPjiAg81-QB>

VGG16 Architectures.

VGG16 is object detection and classification algorithm which is able to classify 1000 images of 1000 different categories with 92.7% accuracy. It is one of the popular algorithms for image classification and is easy to use with transfer learning.



16 Trainable Layers

Conv Layes - Capture Spatial Info

FC Layer - Classification Task

Convolution layers of 3x3 filter with stride 1 and always used the same padding
Maxpool layer of 2x2 filter of stride 2.

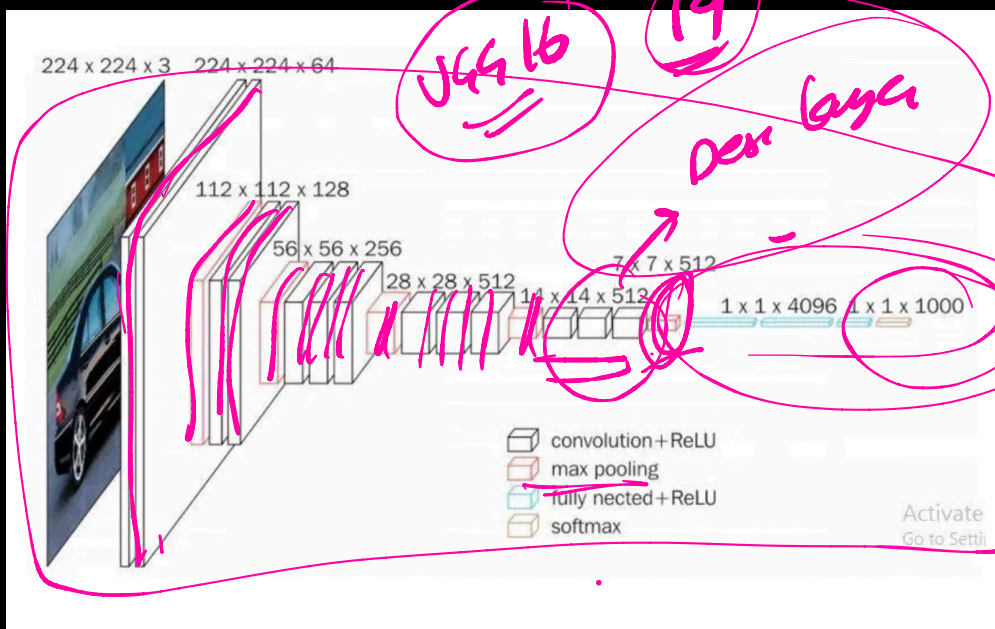
Conv-1 Layer has 64 number of filters

Conv-2 has 128 filters

Conv-3 has 256 filters

Conv 4 and Conv 5 has 512 filters

Three Fully-Connected (FC) layers follow a stack of convolutional layers: the first two have 4096 channels each, the third performs 1000-way ILSVRC classification and thus contains 1000 channels (one for each class). The final layer is the soft-max layer.



Transfer Learning in Action:

1. Freeze all Layers except dense.
2. Fine tune layers with dense.

